

PROJECT REPORT DOCUMENT

Project Title

AI-Based Automatic Waste Classification System Using YOLOv8 Image Classification

Problem Statement

Proper waste segregation is one of the major challenges in modern waste management. Manual classification of waste is time-consuming, prone to human error, and often results in improper disposal, leading to environmental pollution. This project develops an intelligent waste classification system using the YOLOv8 Image Classification model. The system automatically classifies waste images, displays the confidence score, recommends the appropriate disposal method, and presents the results through a React.js dashboard.

Model Explanation

The proposed system uses the YOLOv8 Image Classification model.

Workflow:

1. User uploads or captures a waste image.
2. The React.js frontend sends the image to the Python backend.
3. The backend preprocesses the image.
4. The YOLOv8 model classifies the image.
5. The system predicts the waste category.
6. The confidence score is generated.
7. The disposal method is displayed.
8. The results are shown on the user dashboard.

Technology Stack:

Frontend: React.js, CSS3, JavaScript

Backend: Python (FastAPI)

Model: YOLOv8 Image Classification

Libraries: OpenCV, NumPy, PyTorch

Project Outcome / Results

The system successfully classifies waste into Recyclable, Organic, and Hazardous categories.

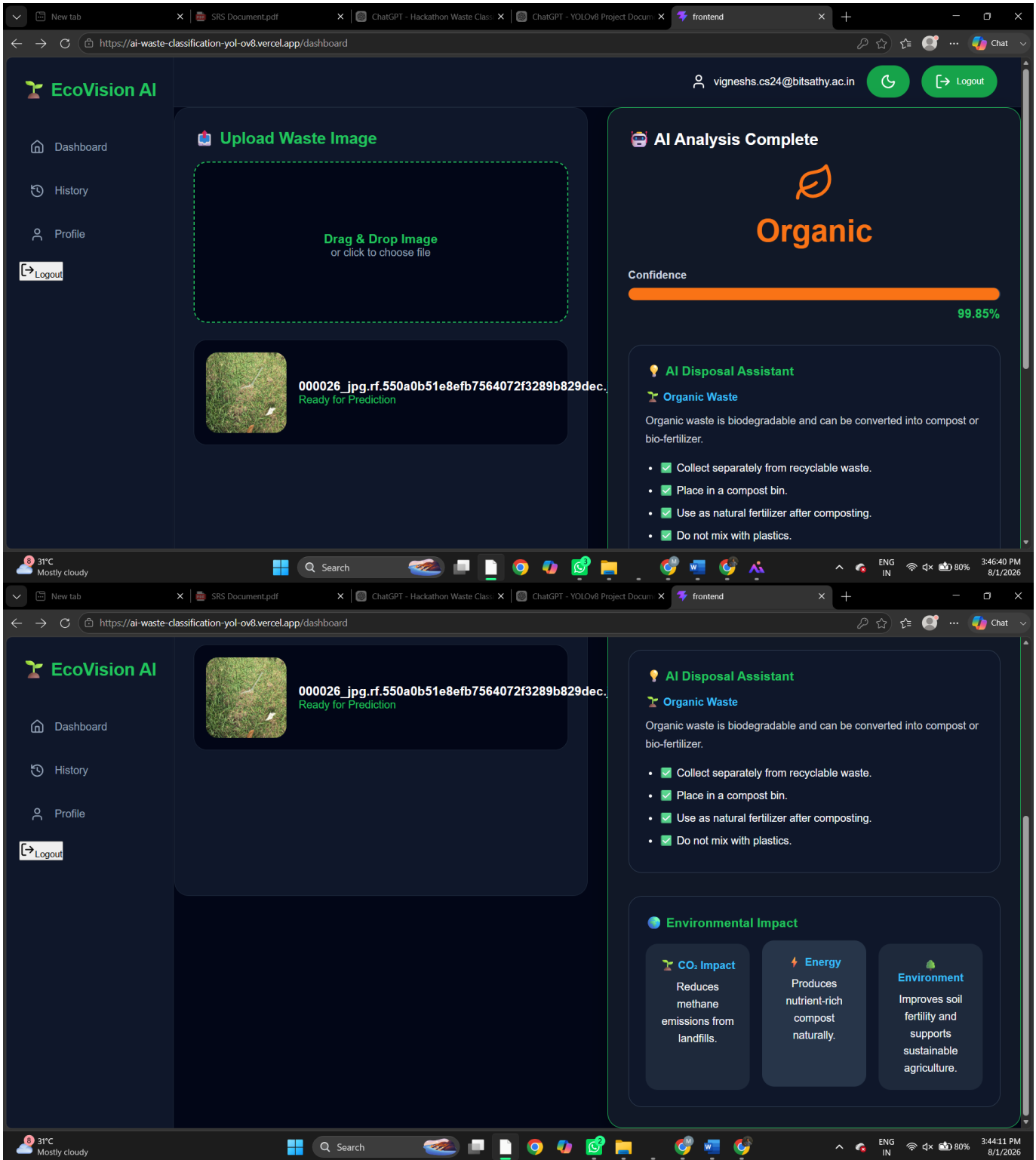
Outputs:

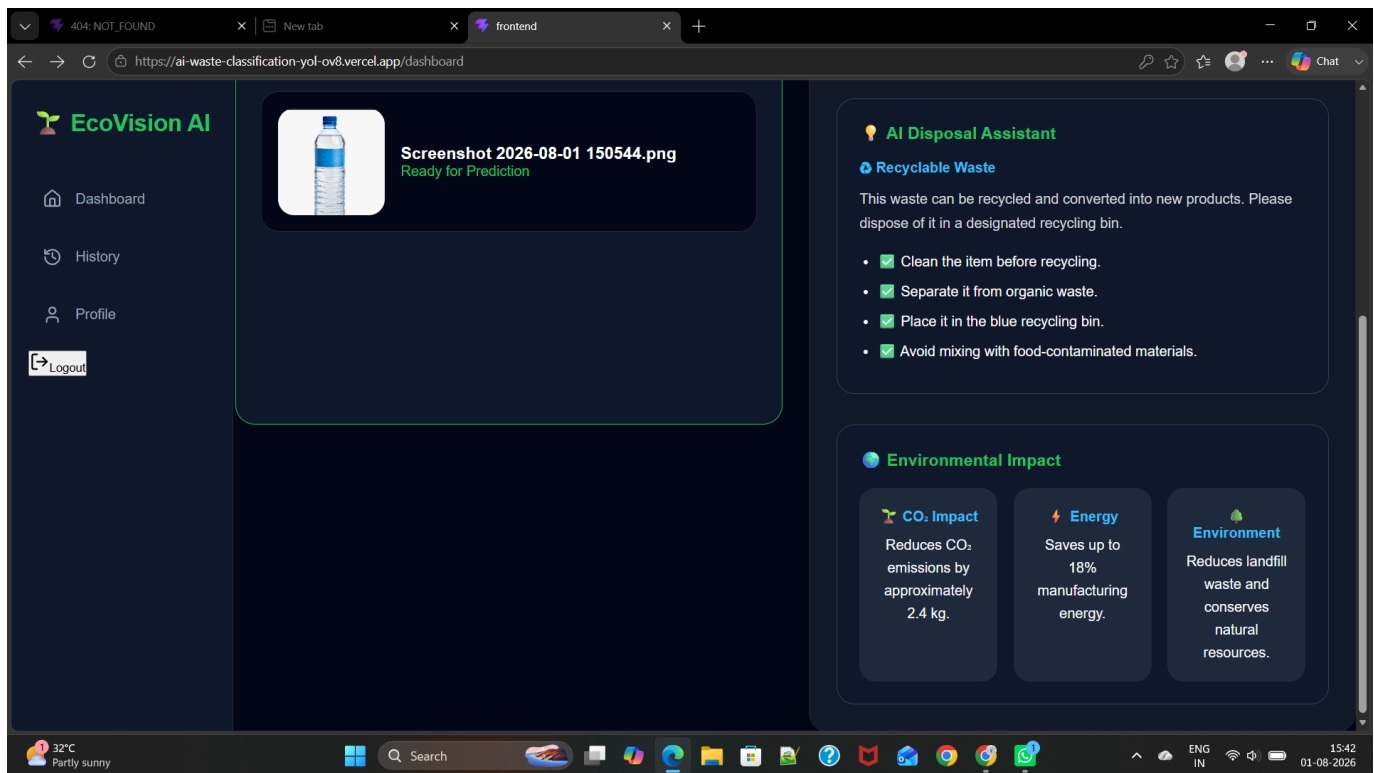
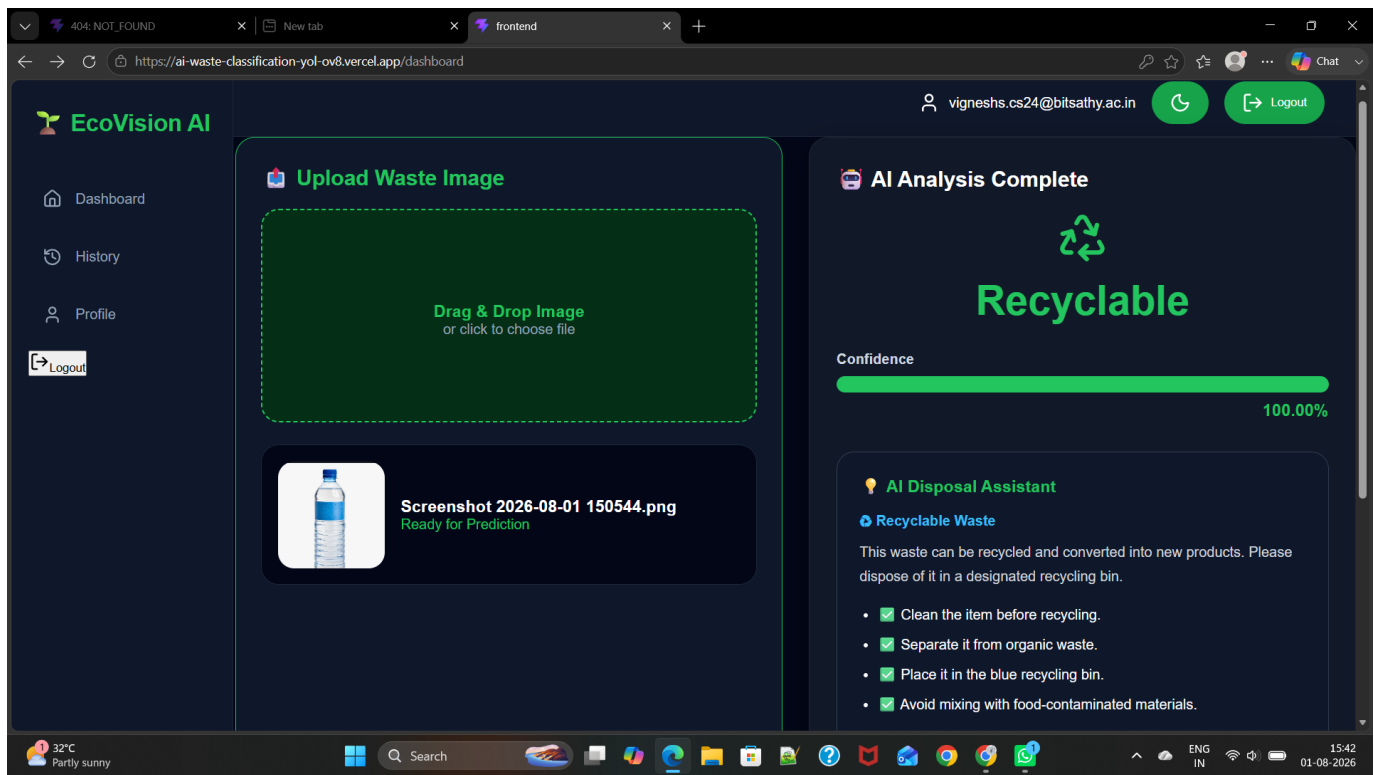
- Waste Category
- Confidence Score
- Disposal Method
- User Dashboard

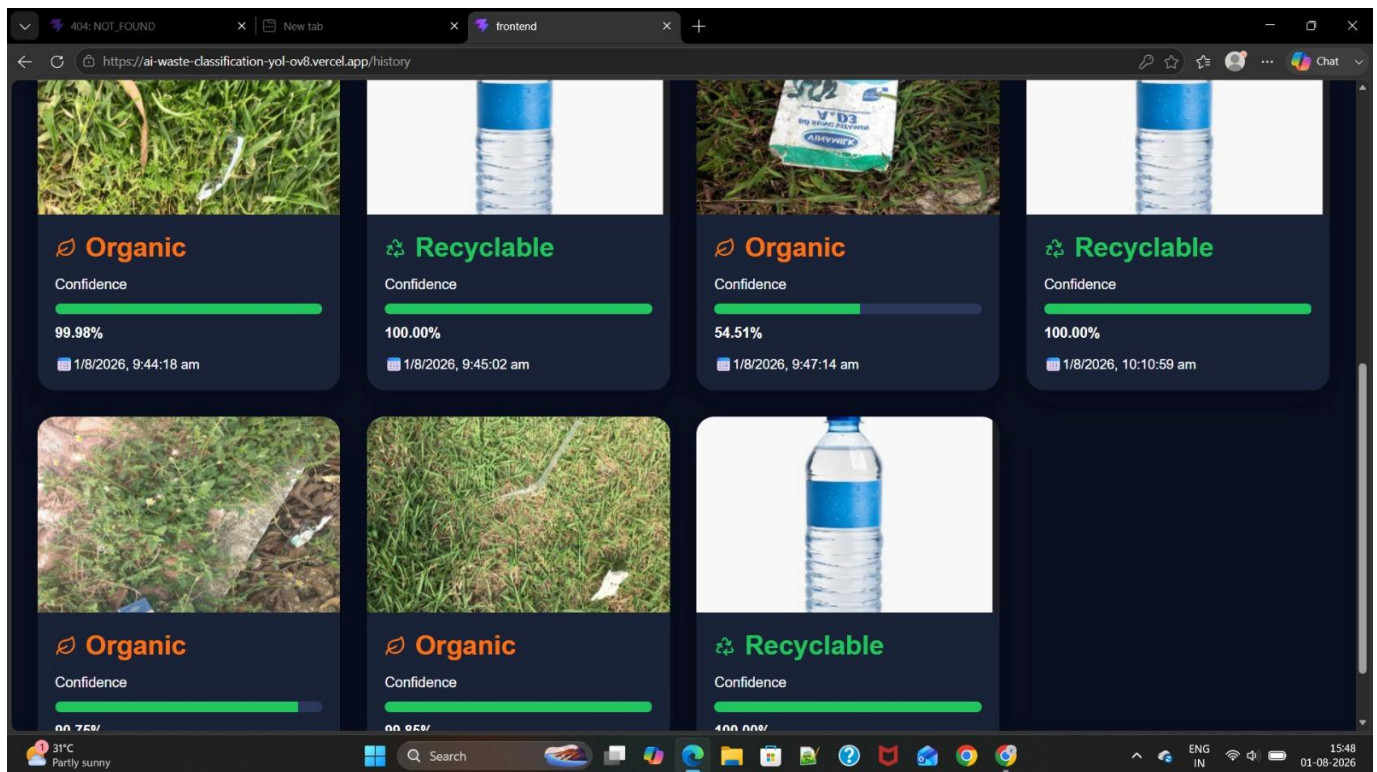
Sample Results:

- Plastic Bottle → Recyclable
- Banana Peel → Organic
- Battery → Hazardous

The application provides fast and accurate predictions while helping users choose the correct disposal method.







Conclusion

The AI-Based Automatic Waste Classification System provides an efficient solution for automated waste segregation using the YOLOv8 Image Classification model. The React.js frontend offers a simple interface, while the Python backend processes images and communicates with the trained model. The project delivers waste classification, confidence scores, disposal recommendations, and a user dashboard, supporting sustainable and intelligent waste management.

GitHub Link : <https://github.com/Selvaganapathy-k/AI-Waste-Classification-YOLOv8.git>